

ReLeaf - Researched Problem Analysis

Manan Sethi - B00949343

Hack the Elements | ShiftKey Labs | May 2026

Team: ReLeaf | Member: Manan Sethi | Theme: Fire + Earth + Air

What is the problem?

After wildfires are extinguished, the damage to soil, vegetation, and air quality persists for months to years, yet no unified tool exists to monitor cross-system ecosystem recovery in real time.

What is happening and why?

Wildfires are increasing in frequency and severity due to climate change. Canada's 2023 season burned over 18 million hectares, the worst on record. Once fires are contained, three interconnected systems begin a slow recovery: soil loses its structure and water retention capacity leading to erosion; vegetation takes 3–15 years to regrow depending on fire severity; and smoke particulates continue degrading air quality long after flames are out. Despite this interconnection, post-fire monitoring treats each system in isolation.

Who is affected?

Rural and Indigenous communities near burn zones facing contaminated water, degraded land, and chronic air quality issues; farmers and landowners dealing with soil erosion and lost productivity; emergency managers lacking unified recovery data; and climate researchers needing longitudinal cross-system data.

What are the gaps in current solutions?

NASA FIRMS tracks active fires but stops at containment. Air quality platforms like OpenAQ don't correlate with burn scars. NDVI vegetation indices exist but aren't accessible to non-technical users. No open tool unifies soil, air, and vegetation recovery into a single interface.

Sources:

Stanford University (May 2024):

<https://news.stanford.edu/stories/2024/05/how-wildfires-change-soil-chemistry>

Earth.Org (February 2026): <https://earth.org/how-ecosystems-recover-after-a-fire/>

Bryant University (January 2026):

<https://news.bryant.edu/year-out-las-devastating-wildfires-bryant-expert-explores-current-state-ecological-recovery>

USGS (December 2025):

<https://www.usgs.gov/data/site-characteristics-postfire-soil-hydrologic-and-biogeochemical-response-and-recovery>

Natural Resources Canada: <https://natural-resources.canada.ca>